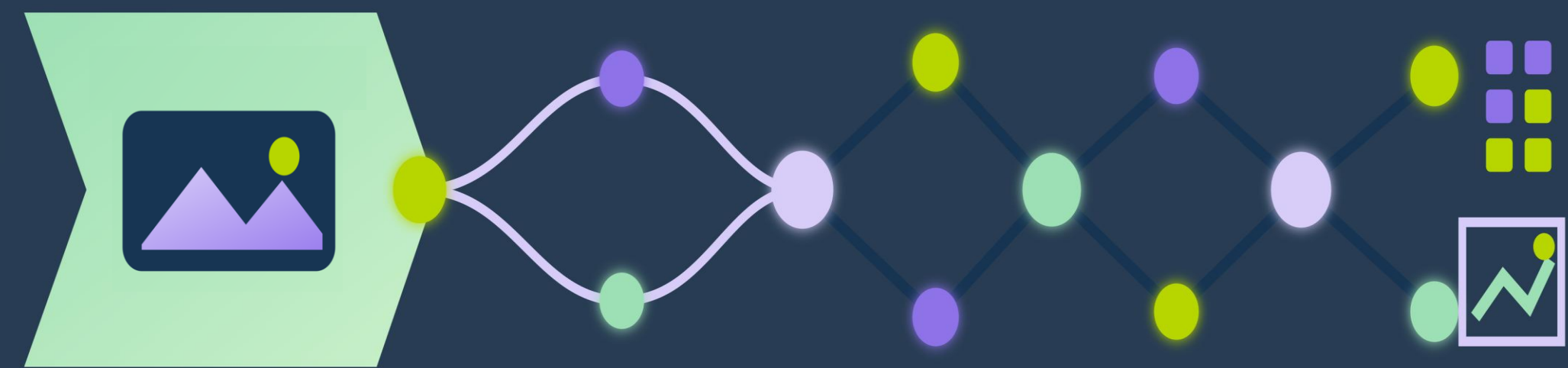


CAN AI DISCOVER EXPLAINABLE FEATURE EXTRACTORS?

An Empirical Investigation in Computer Vision



1. MOTIVATION & QUESTION

Feature extractors produce useful representations—but their decisions can be difficult to inspect or explain.

RESEARCH QUESTION

Can AI-aided search discover compact, explainable feature extractors that generalise to unseen data?

Three explainable search methods are compared with two neural baselines across four image datasets under one evaluation framework.

CAN SEARCH BALANCE ALL THREE?

PERFORMANCE

Does it classify images accurately?

GENERALISATION

Does it work on unseen examples?

EXPLAINABILITY

Can the learned program be inspected?

2. APPROACHES COMPARED

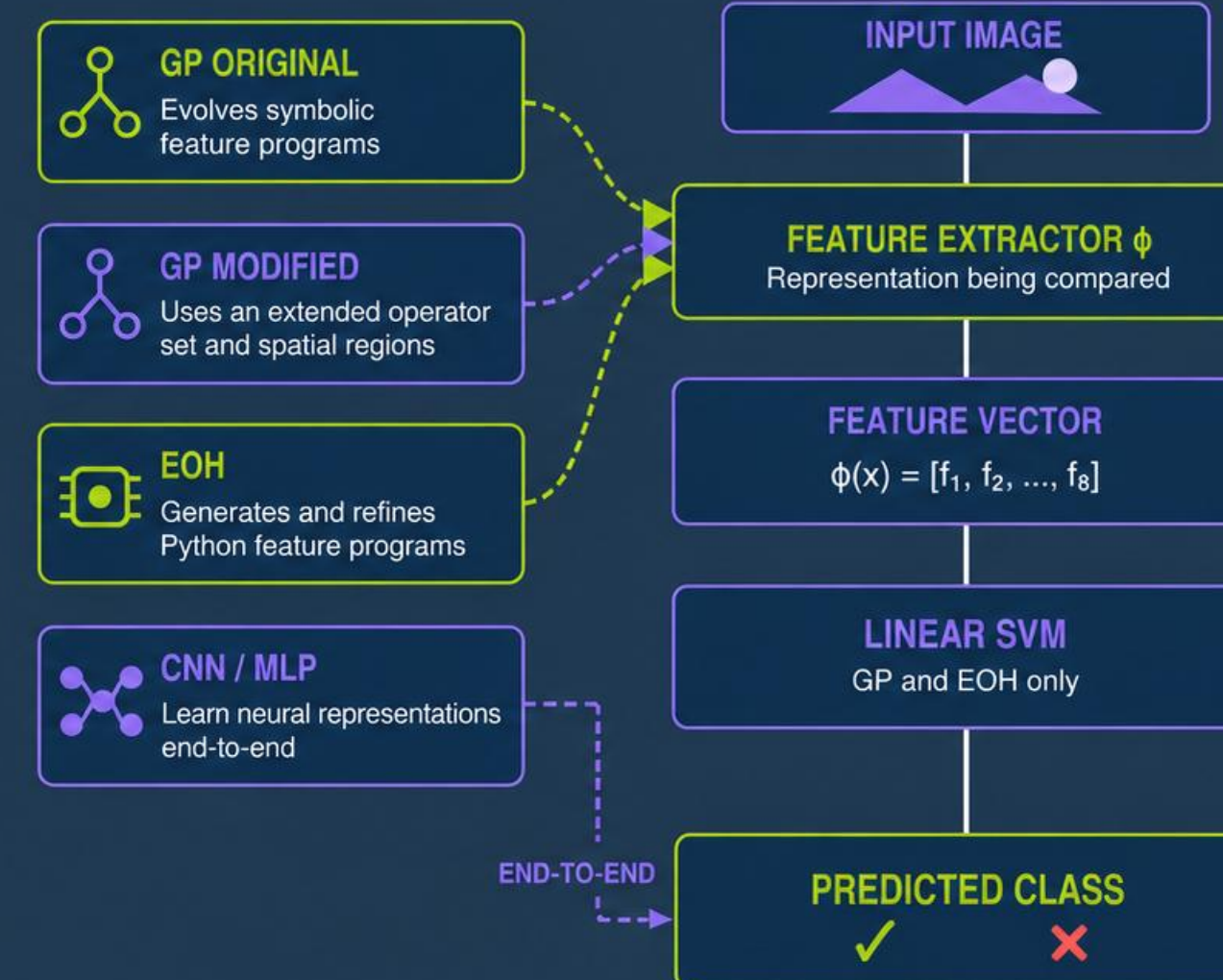
WHAT THE FIVE APPROACHES REPRESENT

Explainable search constructs explicit feature-extraction programs that can be inspected, while neural baselines learn internal representations directly from image pixels.

- Genetic Programming (GP) Original** [1] evolves symbolic feature extractors from established image-processing operators.
- Genetic Programming (GP) Modified** [2] extends this search with additional operators and spatial-region selection.
- Recent LLM-guided approaches show that executable programs can be generated and refined through search [3–5].
- Evolution of Heuristics (EOH)** [4] uses a large language model to generate and refine explicit Python feature-extraction programs.
- Convolutional Neural Network (CNN)** learns spatial image features and predicts classes end-to-end.
- Multilayer Perceptron (MLP)** learns a non-spatial representation directly from flattened image pixels.

COMMON EVALUATION FRAMEWORK

GP and EOH produce explicit feature vectors evaluated by a linear SVM; CNN and MLP learn representations and classify images end-to-end.



All five methods use the same data splits and evaluation metrics.

3. DATASETS & EVALUATION

FOUR CONTRASTING IMAGE DATASETS

The methods were tested across facial, industrial and natural-image classification tasks.



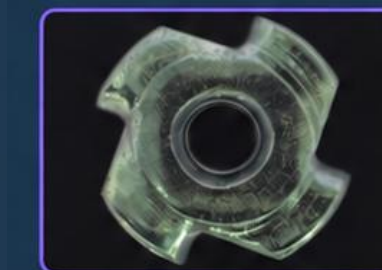
FEI

Face images · identity classification



KSDD2

Steel surfaces · defect detection



MVTec AD

Industrial objects · anomaly detection

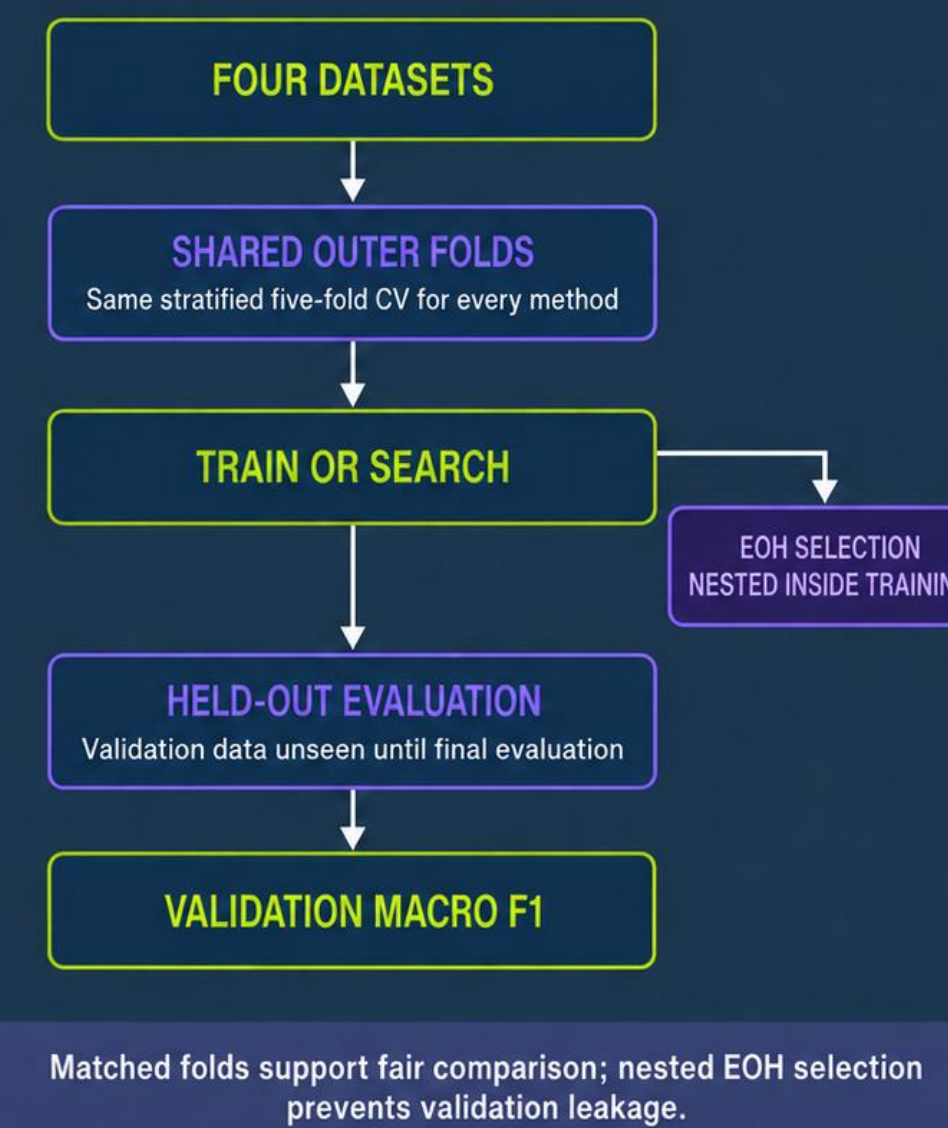


STL-10

Natural RGB images · 10-class classification

MATCHED, LEAKAGE-SAFE EVALUATION

Every method was evaluated on the same held-out folds, enabling direct comparison.

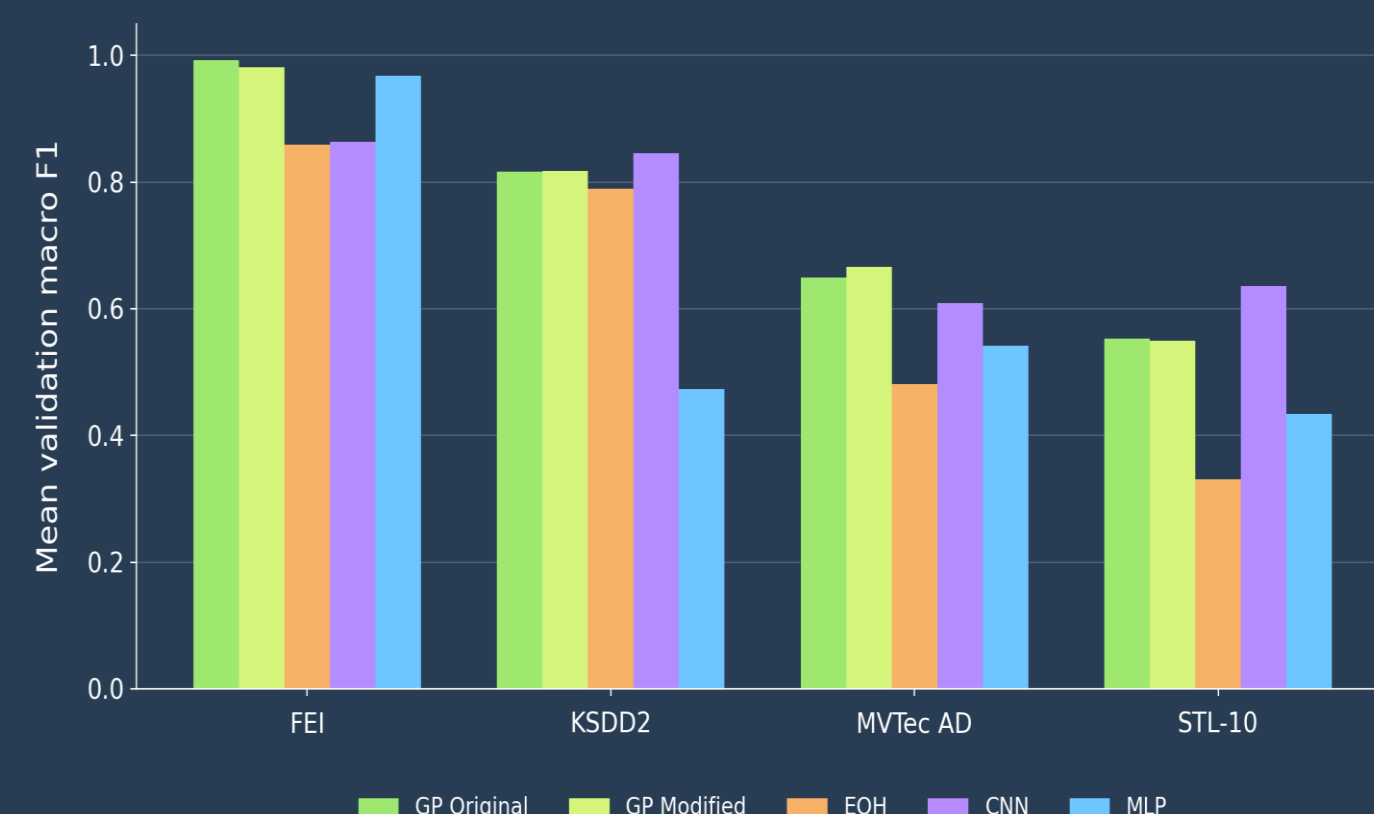


4. WHICH METHODS PERFORMED BEST?

5. EXPLAINING TWO MODELS ON THE SAME IMAGE

6. WHAT IMAGE-LEVEL ANALYSIS REVEALS?

Mean validation macro F1 (seeds 42-43)

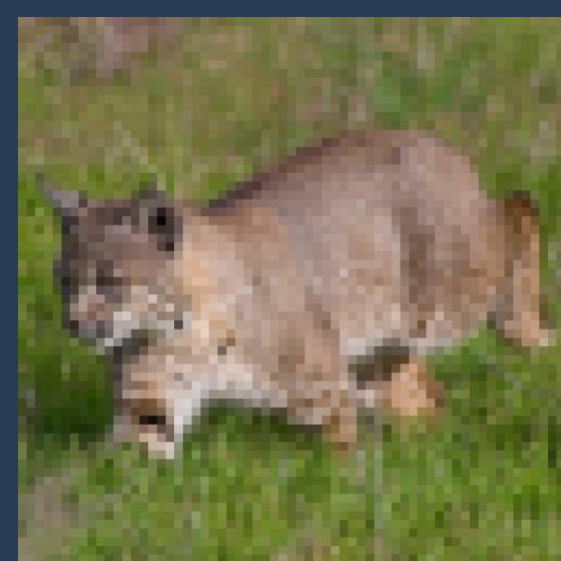


NUMERICAL WINNERS

FEI GP Original 0.993
KSDD2 CNN 0.845
MVTec AD GP Modified 0.666
STL-10 CNN 0.635

No method dominated every dataset. After Holm correction, FEI contained no significant pairwise differences. The clearest separation occurred on KSDD2 and STL-10.

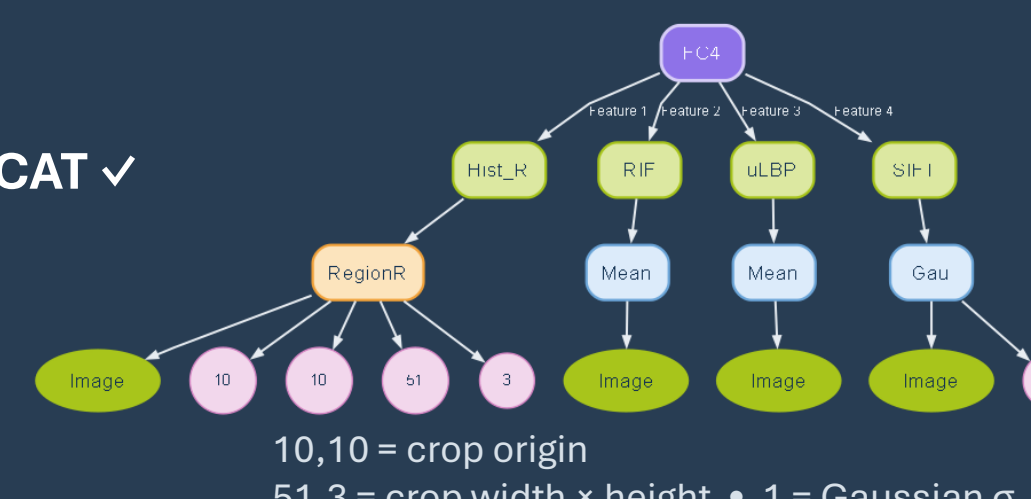
Objectively selected from 69 eligible validation images: GP Modified was correct and EOH failed in both seeds.



TRUE CLASS CAT

GP Modified: CAT ✓
EOH: DEER ✗

GP MODIFIED — 783 FEATURES

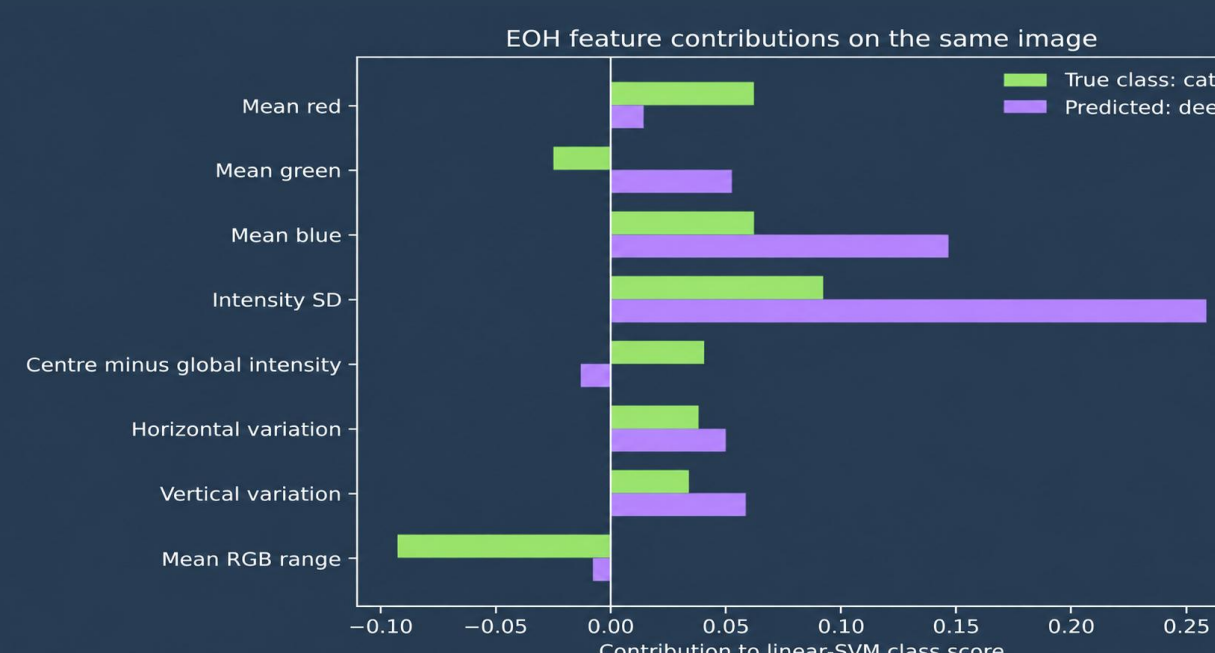


10, 10 = crop origin
51, 3 = crop width × height • 1 = Gaussian σ

EOH - 8 HUMAN-READABLE FEATURES

Mean red, green and blue
Intensity contrast
Central-global intensity
Horizontal/vertical variation
Mean RGB colour range

Exact standardised-feature × linear-SVM weight contributions. They describe the fitted model, not causal importance.



Same image, different representations: GP combined regional colour, texture and SIFT descriptors; EOH remained compact but supported the wrong class.

36,404

seed-level observations
18,202 underlying images

ALL 8 DATASET × SEED BLOCKS PASSED
0 missing keys • 0 extra keys • 0 identity mismatches

STL-10 COMPLEMENTARY SUCCESSES

2,466

observations where
CNN was correct while
both GP and EOH failed

4,474

observations where at
least one GP method
was correct while CNN
failed

Aggregate scores conceal image-specific strengths and failures. Interpretable feature search and neural models often solve different cases—even when their overall scores are similar.

NO SINGLE REPRESENTATION WINS EVERYWHERE

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[2] Y. Bi *et al.*, *IEEE TEVC*, 25(1):87–101, 2021.
[3] B. Romera-Paredes *et al.*, *Nature*, 625:468–475, 2024.

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GitHub: Scan the QR code to view the project repository and code.

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PROJECT CODE & FURTHER ANALYSIS:

